Nhat Minh Doan (Đoàn Nhật Minh)

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Research area Hyperbolic geometry and low-dimensional topology. My current research concerns the multiplicity of simple closed geodesics on hyperbolic surfaces.

Education

PhD in Mathematics, University of Luxembourg, Luxembourg — 2018 - 2022
Advisor: Prof. Hugo Parlier
Thesis: Quantifying some properties of curves and arcs on hyperbolic surfaces

MSc in Mathematics, University of Nice Sophia Antipolis, France — 2016 - 2017
Advisor: Prof. François Labourie
Thesis: Arithmetic and ergodic theory of hyperbolic surfaces

BA in Mathematics, University of Education, Ho Chi Minh City, Vietnam — 2013 - 2015

Academic positions

[3] **Researcher**, Institute of Mathematics, VAST, Vietnam — 2022 - present

[2] **SASEA Research Fellow**, National University of Singapore, Singapore — 2024 - 2026
Advisor: Prof. Ser Peow Tan

[1] **Doctoral Researcher**, University of Luxembourg, Luxembourg — 2018 - 2022

Papers

[11] Density of length ratios on Teichmüller space — 2026
(with Hanh Vo)
arXiv:2610.24700.

[10] Stable-norm boundary geometry on once-punctured tori — 2026
(with Xiaobin Li and Van Nguyen)
arXiv:2605.14574. Submitted.

[9] Graded Bridgeman dilogarithm identities on hyperbolic surfaces — 2026
(with Ara Basmajian, Hugo Parlier and Ser Peow Tan)
arXiv:2601.03876. Submitted.

[8] Optimal Farey sequence for the congruence subgroup $\Gamma_0(2^n)$ — 2026
(with Sang-hyun Kim, Mong Lung Lang and Ser Peow Tan)
arXiv:2601.01324. Submitted.

[7] Some arithmetic aspects of ortho-integral surfaces — 2026
(with Khanh Le)
Trans. AMS, 379 (2026), 6365 - 6392. arXiv:2504.09403

[6] Some remarks on subgroups of free-by-cyclic groups — 2026
(with Hoang Thanh Nguyen)
Vietnam J. Math., 54, 715 - 728 (2026).

[5] Self-intersections of arcs on a pair of pants — 2025
(with Hanh Vo)
Bull. LMS, 57 (2025), 2190 - 2209. arXiv:2407.17750

[4] Optimal special polygons for the congruence subgroups $\Gamma_0(p)$ and $\Gamma_0(pq)$ — 2025
(with Sang-hyun Kim, Mong Lung Lang and Ser Peow Tan)
J. Geom. Anal., Volume 35, 216 (2025). arXiv:2209.13937

	[3] Ortho-integral surfaces Adv. Math., Volume 466, April 2025, 110162. arXiv:2112.10694	2025
	[2] Geometric filling curves on punctured surfaces Glasgow Math. J., Volume 65, Issue 2, May 2023, pp. 383 - 400. arXiv:2012.02461	2023
	[1] Measuring pants (with Hugo Parlier and Ser Peow Tan) Trans. AMS, 376 (2023), 5281 - 5306. arXiv:2002.02738	2023
Other writings	[1] Almost all simple lengths are unique up to isometry hexagon:2610.00298v1.	2026
	[2] The torus with asymptotically the fewest simple closed geodesics (with Phuc Thinh Dang and Trong Toan Dao) hexagon:2610.00159v1.	2026
Referee work	International Mathematics Research Notices; Proceedings of the American Mathematical Society; Mathematische Annalen; Transactions of the American Mathematical Society; Conformal Geometry and Dynamics.	
Talks	[28] McShane birthday conference, Hebei Normal University, China	Aug-Sep 2026
	[27] Seminar, TIMAS, Thang Long University, Vietnam	June 2026
	[26] Topology and Geometry Seminar, NUS, Singapore	May 2026
	[25] Topology and Geometry Seminar, Southwest Jiaotong University, China	March 2026
	[24] Workshop on Geometry and related topics, Phu Tho, Vietnam	Jul 2025
	[23] Grad Student Seminar, NUS, Singapore	Apr 2025
	[22] Teichmüller Theory and Flat Structures, Matrix, Creswick, Australia	Mar 2025
	[21] Workshop on Dynamics and Related Topics, IMS, NUS, Singapore	Mar 2025
	[20] Workshop on Geometry and Dynamics in Low Dimensions, IMVAST, Hanoi	Jan 2025
	[19] Scientific workshop for young researchers, Vinh Phuc, Vietnam	Sep 2024
	[18] IMVAST-OCAMI Joint Conference, Osaka Metropolitan University, Japan	May 2024
	[17] Geometry and Topology Seminar, KIAS, Korea	Mar 2024
	[16] Workshop on Hyperbolic Geometry of Numbers, HCMC, KIAS, Korea	Dec 2023
	[15] JSPS-VAST Workshop, FPT University Quy Nhon, Vietnam	Sep 2023
	[14] Mathematics workshop and related fields, Ha Tinh University, Vietnam	Apr 2023
	[13] JSPS-VAST Bilateral Joint Research Project Workshop, IMVAST	Dec 2022
	[12] Algebra and Number Theory Seminar, IMVAST	Nov 2022
	[11] Scientific conference on Geometry, Analysis and Applications, Phu Tho, Vietnam	Oct 2022
	[10] Topology and Geometry Seminar, NUS, Singapore	Sep 2022
	[9] Scientific workshop for young researchers, Vinh Phuc, Vietnam	Aug 2022
	[8] Japan-Vietnam Singularity Seminar (Online), IMVAST	Apr 2022
	[7] Mostly Teichmüller Spaces Seminar (Online), Queens College, USA	Mar 2022
	[6] Ph.D. Defense, Geometry Winter Workshop, Luxembourg	Jan 2022
	[5] Topology and Geometry Seminar (Online), IMVAST	Nov 2021
	[4] Topology and Geometry Seminar (Online), VNU, Ho Chi Minh City, Vietnam	Oct 2021
	[3] PhD Away Days (Online), University of Luxembourg, Luxembourg	Oct 2020
	[2] PhD Seminar (Online), University of Luxembourg, Luxembourg	May 2020

	[1] School on Geometry and Dynamics on Hyperbolic Surfaces, Switzerland	Aug 2018
Services	[3] Co-organizing Geometry and Dynamics in Low Dimensions Workshop, Hanoi With S. Kim, T. Koberda, C. Leininger, S. P. Tan, T. T. Nguyen, T. D. Do, and V. T. Tran	Jan 2025
	[2] Co-organizing Geometry, Topology & Dynamics Seminar, National University of Singapore With Tengren Zhang and Daren Wei	2024 - 2026
	[1] Co-writing a short comic story: Maths, Magic and Maryam	2020
Advising	[3] Undergraduate student: Phuc Thinh Dang	2026 - present
	[2] PhD student: Van Nguyen	2025 - present
	[1] Vietnamese REU students: Trong Toan Dao and Anh Tuan Nguyen Co-supervised with Ser Peow Tan	2023 - 2024
Teaching	[5] Modern Geometry, Master program, Institute of Mathematics, VAST	2023
	[4] Probability, Applied Mathematics and Data Science students, Vietnam France University	2022
	[3] Linear Algebra (teaching assistant), University of Engineering and Technology	2022
	[2] Analysis for Applications (teaching assistant), University of Luxembourg	2018 - 2020
	[1] Elementary Mathematics, children aged 5-10, Mathnasium Vietnam	2012 - 2013
Awards	[1] Excellent Doctoral Thesis Award, University of Luxembourg	2022
Grants	[5] Singapore Academies South-East Asia Fellowship, hosted at the National University of Singapore	2024 - 2026
	[4] NAFOSTED, Vietnam (2 years)	2024 - 2026
	[3] Research Grant, Institute of Mathematics, VAST, Vietnam (2 years)	2022 - 2024
	[2] PRIDE Grant, Fonds National de la Recherche, Luxembourg (4 years)	2018 - 2022
	[1] Master 2 Excellence Scholarship (10 months) J. A. Dieudonné Mathematics Laboratory, Nice, France	2016 - 2017

References

Prof. Ara Basmajian
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Prof. Hugo Parlier
University of Fribourg
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Prof. Ser Peow Tan
National University of Singapore
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Hobbies Table tennis, badminton, hiking